

有机化学(B)中期考试试卷

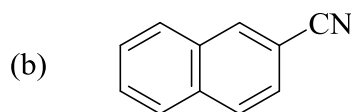
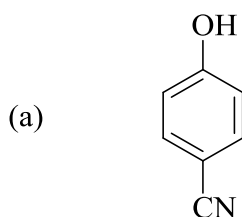
授课老师:

请同学在空白纸上写答案，不用抄写题目

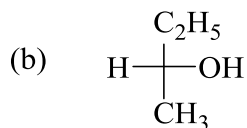
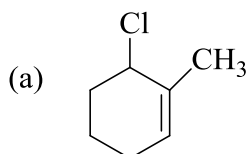
(考试时间: 100 分钟)

一、请回答下列问题：（26 分）

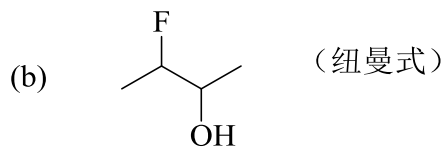
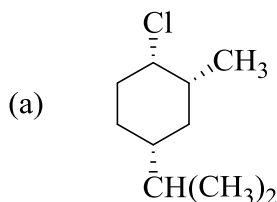
1. 用箭头表明下列化合物硝化时导入硝基的位置 (4 分)。



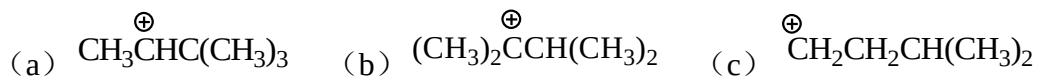
2. 用系统命名法命名下列化合物 (4 分)。



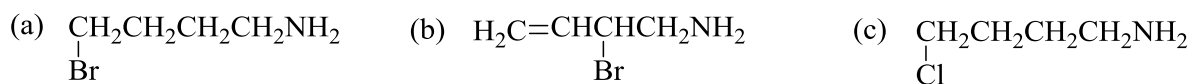
3. 写出下列化合物最稳定的构象式 (4 分)。



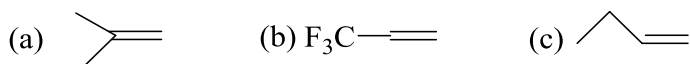
4. 将下列碳正离子按稳定性由大到小排序 (2 分)。



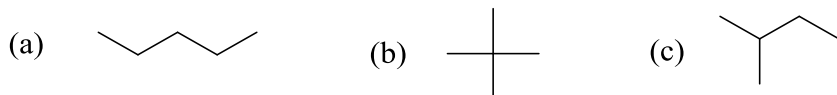
5. 将下列化合物按其发生 S_N1 反应的活性由高到低排序 (2 分)。



6. 将下列化合物按其与浓硫酸的反应活性由高到低排序 (2 分)。



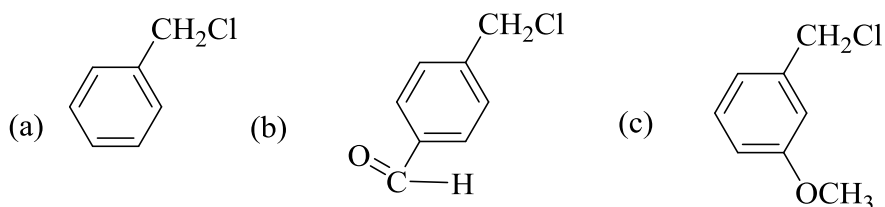
7. 将下列化合物按沸点由高到低排序 (2 分)。



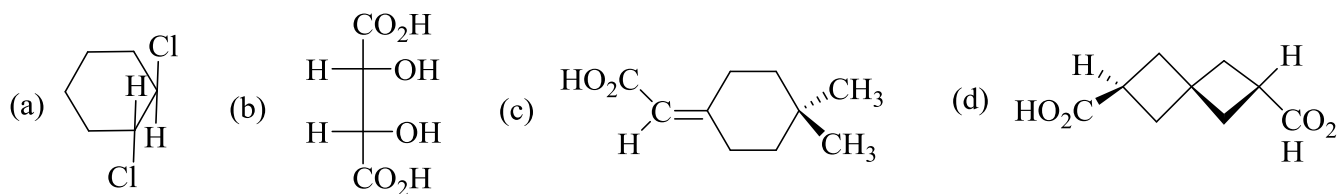
8. 将下列化合物按其发生 S_N2 反应的活性由高到低排序 (2 分)。



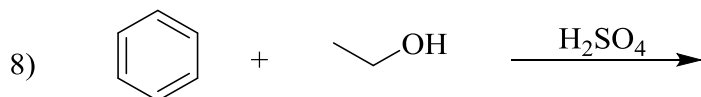
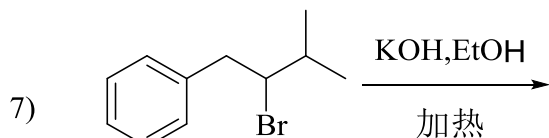
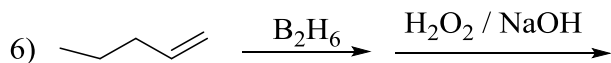
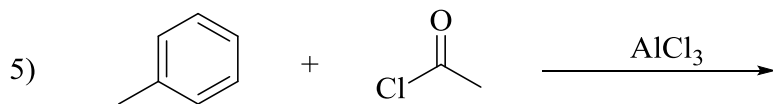
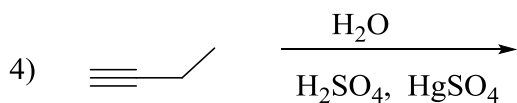
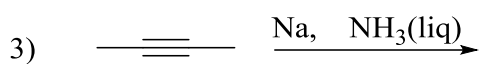
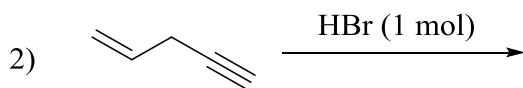
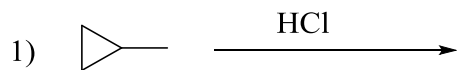
9. 将下列化合物与 $(\text{CH}_3\text{COCl}) / \text{AlCl}_3$ 反应的活性由高到低排序 (2 分)。

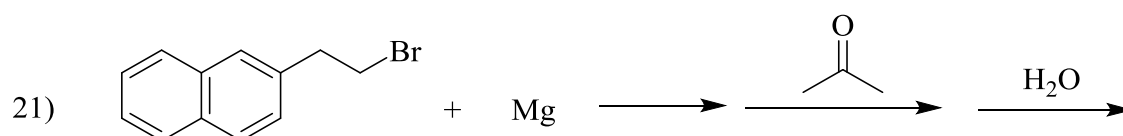
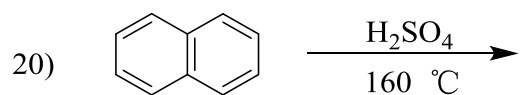
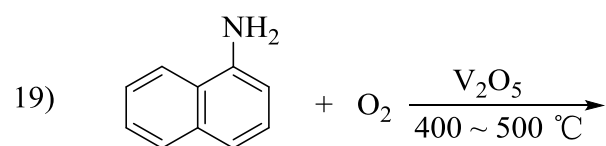
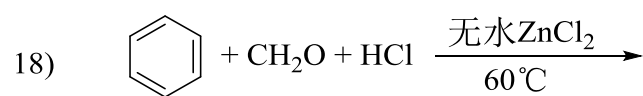
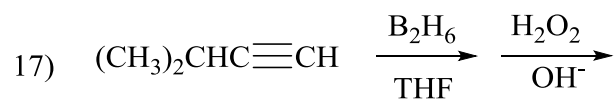
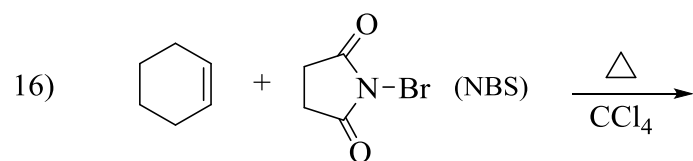
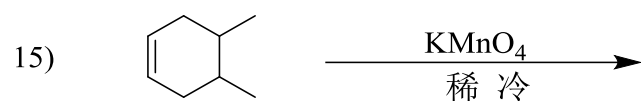
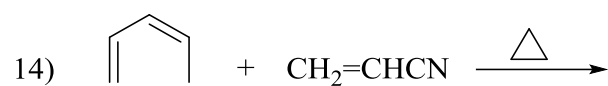
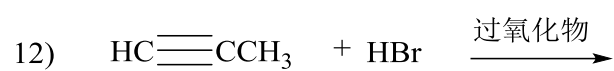
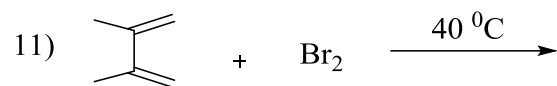
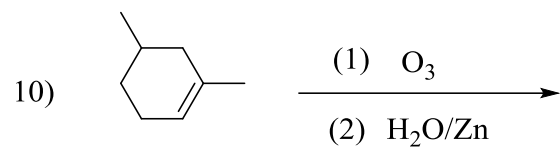
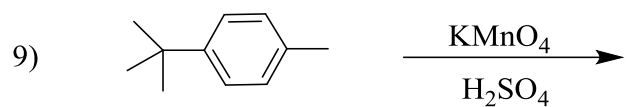


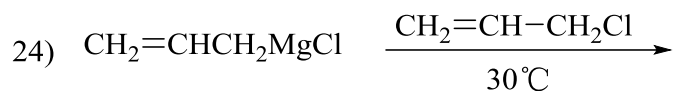
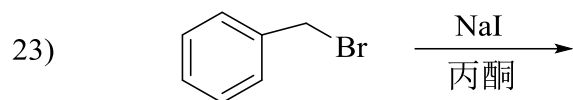
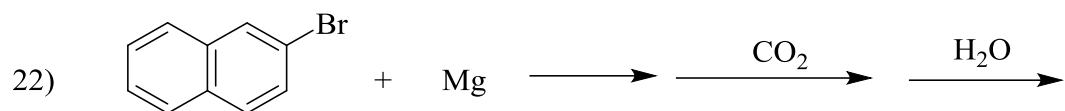
10. 下列哪些化合物具有手性 (2 分)。



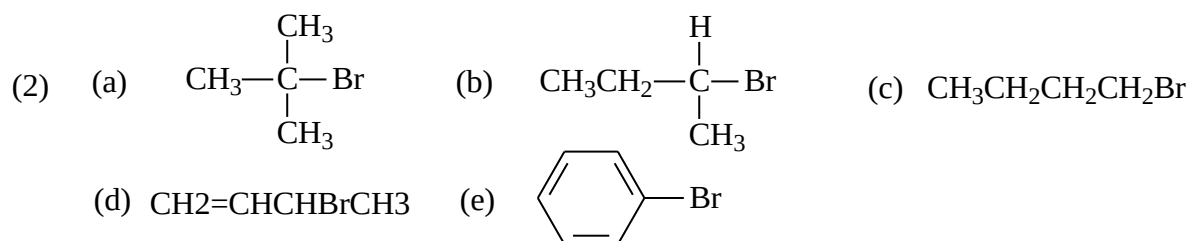
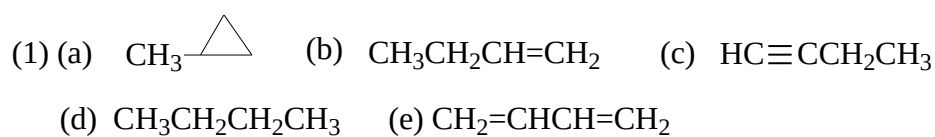
二、请写出下列反应的主要产物 (每题 1 分, 必要时注明立体化学问题, 共 24 分)





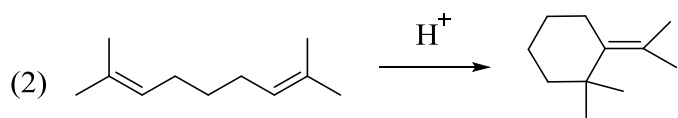
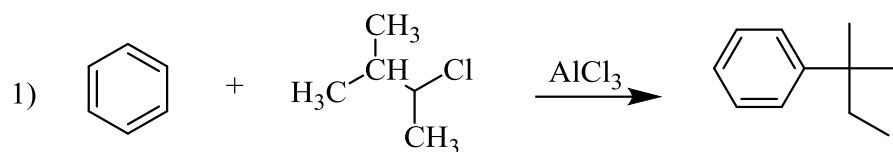


三、请用化学法鉴别下列化合物（10 分）



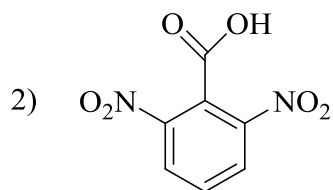
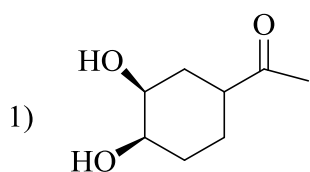
四、机理题(共 2 小题，10 分)。

（请写出下列反应的机理，用箭头表示电子对的转移，鱼钩箭头表示单电子的转移。）

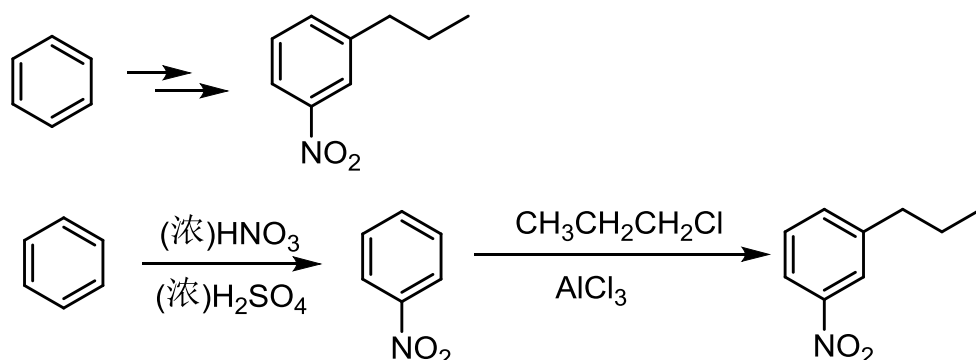


五、合成题（共 2 小题，10 分）。

请用苯、甲苯及 C3 以下的有机化合物(包括 C3)为起始原料合成下列化合物(无机试剂任选)。



六、为了实现以苯为原料制备 3-硝基丙基苯（反应温度、溶剂条件不用分析），有同学提出了如下合成方案。1) 请你找出其中可能的错误并说明错误原因；2) 请你写出苯为原料制备 3-硝基丙基苯的合理路线（每个问题 5 分，共 10 分）。



七、为了实现以两个碳或一个碳为原料制备如下产物（反应温度、溶剂条件不用分析），有同学提出了如下的合成方案。1) 请你找出（I-IV）步骤中可能的错误并说明错误原因；2) 请你以两个碳或一个碳为原料制备如下产物，写出合理反应路线（每个问题 5 分，共 10 分）。

